CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (All Branches)

December 2012

CL103.01 Mechanics of Solids

Date: 07.12.2012, Friday	Time: 01:30 p.m. To 04:30 p.m.	Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Determine the forces in all members of a truss shown in **Fig. 1**. [07]

Q - 2 (a) A uniform ladder of weight 250N and length 5m is placed against a vertical wall in a position where its inclination to the vertical is 30^0. A man weighing 800N climbs the ladder. At what position, will he induce slipping? Take co-efficient of friction as 0.2 at both the contact surfaces of the ladder. [07]

(b) **Fill up the blanks. (Any Seven)** [07]

(i) A body which does not deform under the action of applied forces is called ________.

(ii) Magnitude, direction, sense and ________ are the characteristics of a force.

(iii) The splitting of a force into two perpendicular directions without changing its effect is called ________.

(iv) The force equal and opposite to the resultant of several forces acting on the body is called ________.

(v) The moment of a couple is known as ________.

(vi) The rotational tendency of a force is called ________.

(vii) The CG of a solid circular cone divides the axis in the ratio ________.

(viii) Frictional force is ________ of area of contact.

(ix) Dynamic friction is also known as ________.

Q - 3 (a) Locate the position of the centroid of the plane shaded area depicted in **Fig. 2**. [07]

(b) **Attempt any One** [07]

(i) Find resultant of forces shown in **Fig. 3**.

(ii) Three forces equal to 1kN, 2kN and 3kN are respectively acting in order along the three sides of an equilateral triangle. Make calculations for the magnitude, direction and position of their resultant.

SECTION – II

Q - 4 Enlist six mechanical properties of a material useful from engineering aspect. Explain any two in detail. [07]

Q -.5 (a) Derive an expression for deformation of a uniformly tapered section $\delta l = 4PL/\Pi E d_1 d_2$ with usual notations. [04]

(b) **Attempt any Two** [10]

(i) A stepped bar is loaded as shown in **Fig. 4.** Calculate the stresses in each part and total change in the length of the bar. Take Esteel = 200 GPa, Ecopper = 100 GPa and Ebrass = 80 GPa.

(ii) A steel rod of 100mm dia is inserted into a copper tube of 200mm external diameter and 100mm internal diameter. The composite section is subjected to axial tensile force of 100kN. Length of the section is 0.5m. Esteel = 210 GPa, Ecopper = 130 GPa. Calculate stresses in each materials.

(iii) A steel rod of cross sectional area 4 cm^2 connects two parallel walls 6m apart. When rod is heated to 100°C, the nuts at the ends are tightened up. Calculate the pull exerted by the bar on cooling to 20°C. $\alpha s = 0.000015/^{o}C$, $Es = 2*10^5$ N/mm^2.

Q - 6 Draw shear force and bending moment diagrams indicating values at all important points for any **two beams** from the beams shown in **Fig. 5, Fig. 6 & Fig. 7.** Locate point of contra-flexure if any. **[14]**

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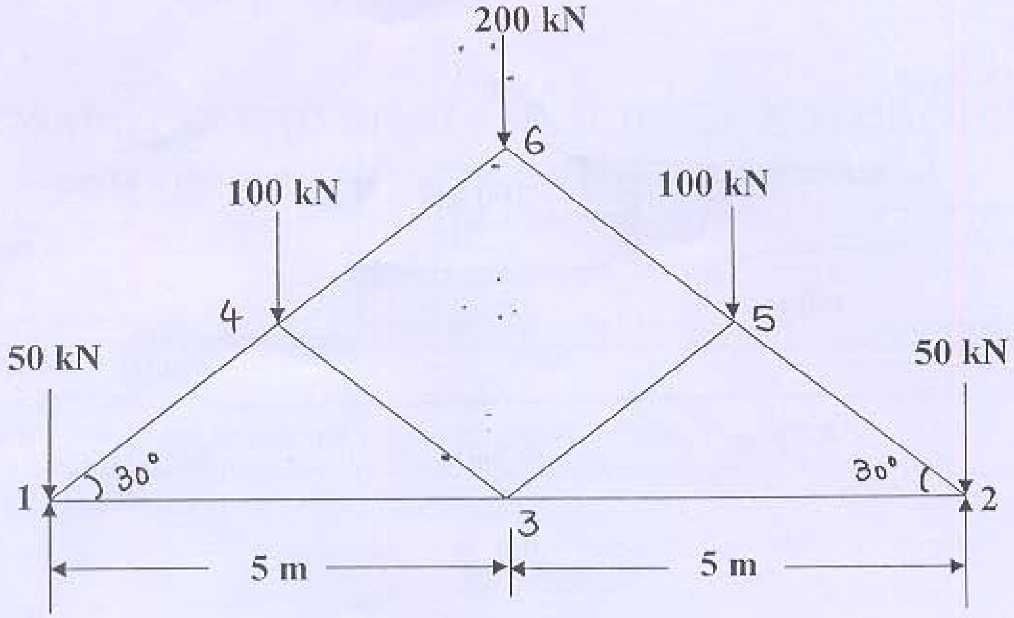

Fig. 1

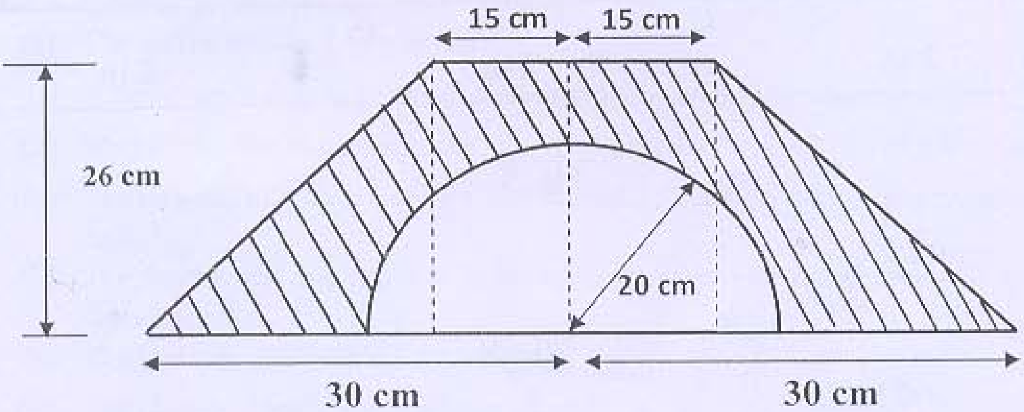

Fig. 2

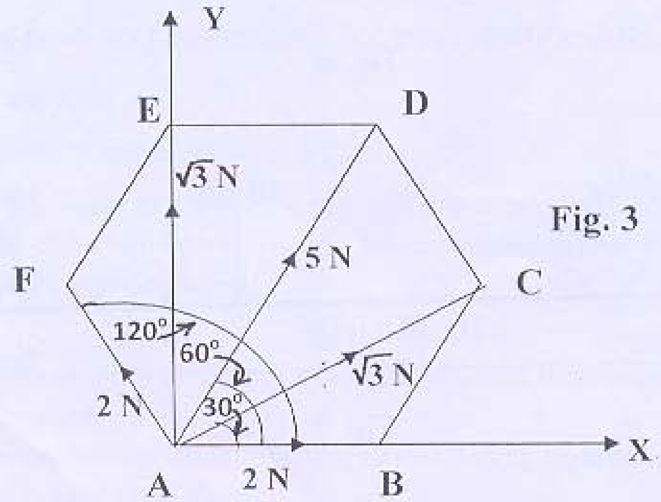

Fig. 3

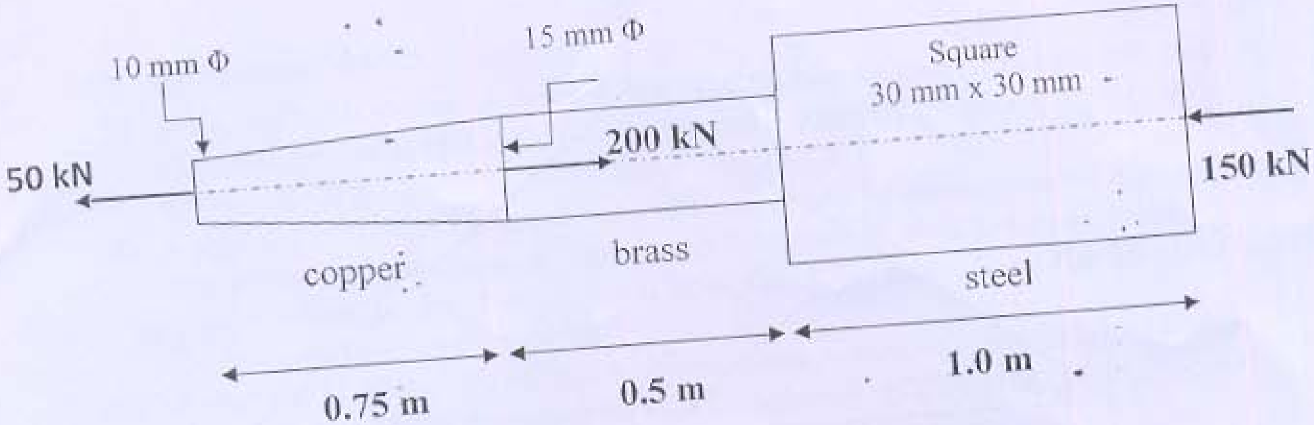

Fig. 4

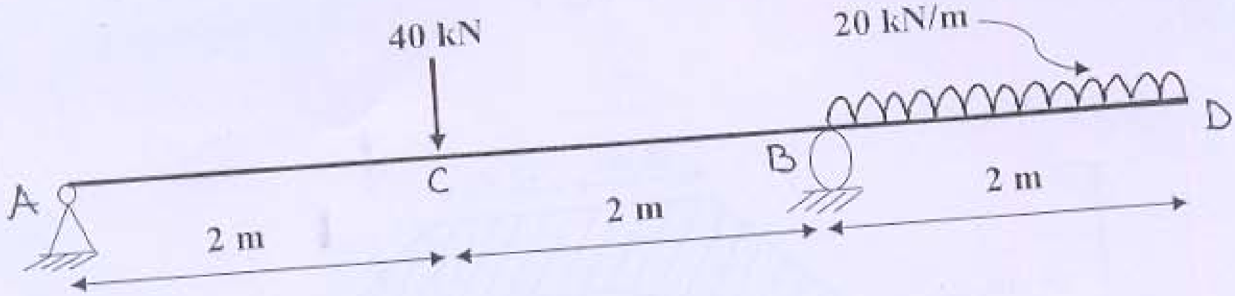

Fig. 5

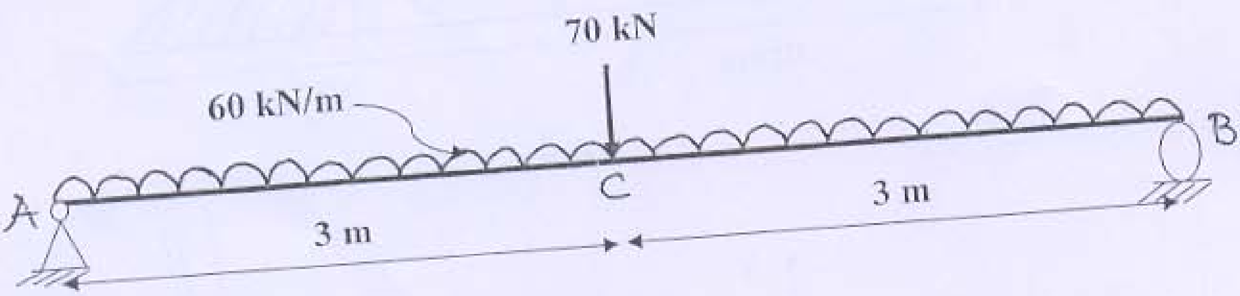

Fig. 6

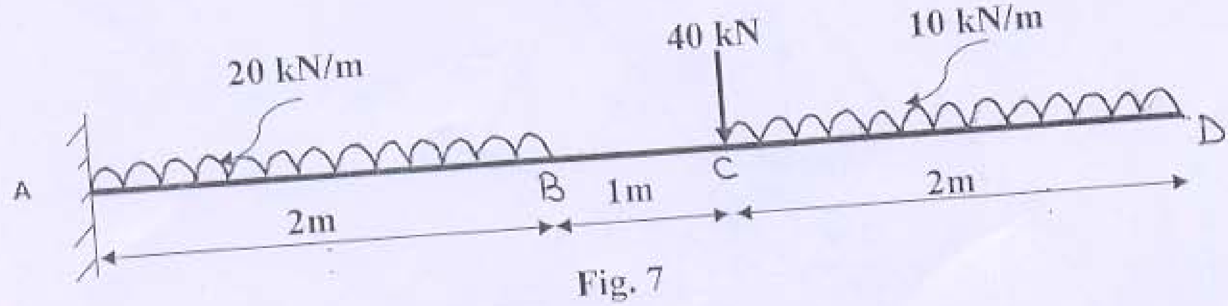

Fig. 7